Technical Report: GloboGym Gym Management System

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1. Introduction

The rapid growth of fitness centers has increased the need for efficient software solutions to manage daily operations such as membership handling, scheduling, and payments. Manual systems are often inefficient, error-prone, and difficult to scale.

This project presents **GloboGym**, a Java-based desktop application developed using **JavaFX**. The system automates essential gym management processes while providing an intuitive graphical user interface for different user roles, including members, coaches, and managers.

2. Project Objectives

The main objectives of the GloboGym project are:

- To design a desktop application that manages gym operations efficiently
- To implement role-based access control
- To support scheduling and booking of gym sessions
- To ensure data persistence using serialization
- To apply object-oriented programming principles and UML modeling

3. Problem Statement

Traditional gym management often relies on manual record-keeping or fragmented digital tools. This leads to:

- Data inconsistency
- Scheduling conflicts
- Poor user experience
- Lack of persistent data storage

GloboGym addresses these issues by providing a centralized, automated, and user-friendly desktop application.

4. System Requirements

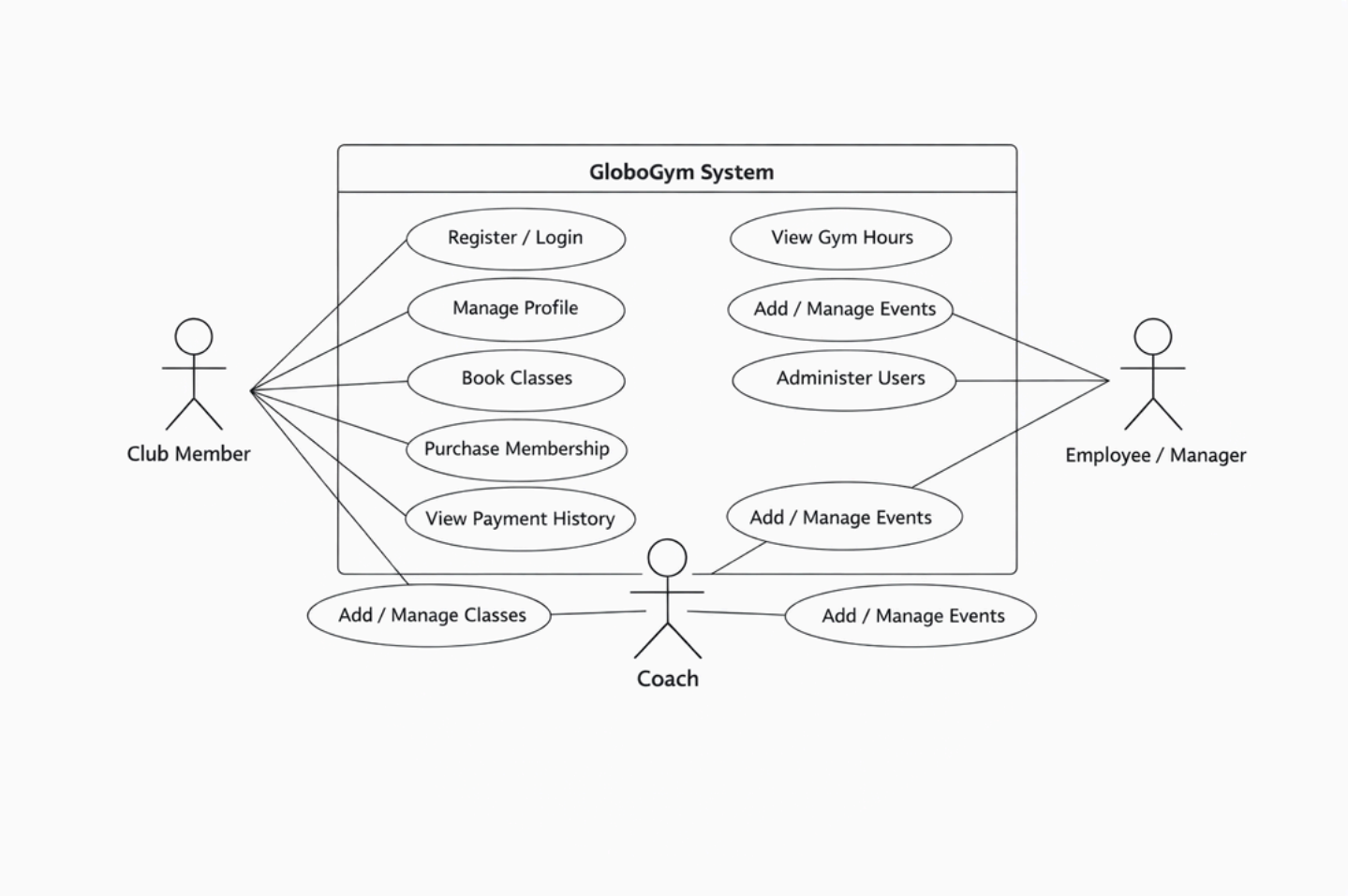
4.1 Functional Requirements

- User registration and authentication

- Role-based access (Member, Coach, Manager)
- Membership management
- Class scheduling and booking
- Payment tracking
- Persistent storage of all data

4.2 Use Case Diagram

Figure 1: Use Case Diagram of GloboGym



The use case diagram illustrates the interactions between users and the system. Members can register, log in, book classes, and make payments. Coaches manage schedules, while managers oversee memberships and system operations. This diagram provides a high-level view of the system's functional behavior.

5. System Design

5.1 Architecture Overview

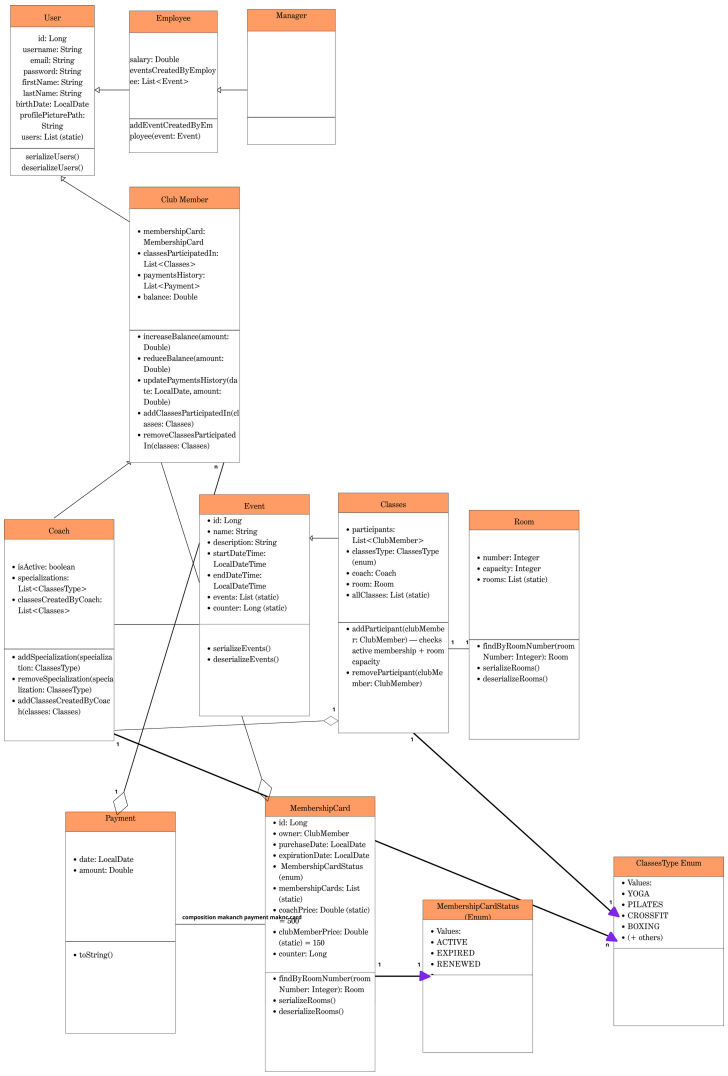
GloboGym follows a modular architecture based on object-oriented principles. The system separates concerns between:

- User interface (JavaFX)

- Business logic
- Data persistence using serialization

5.2 Class Diagram

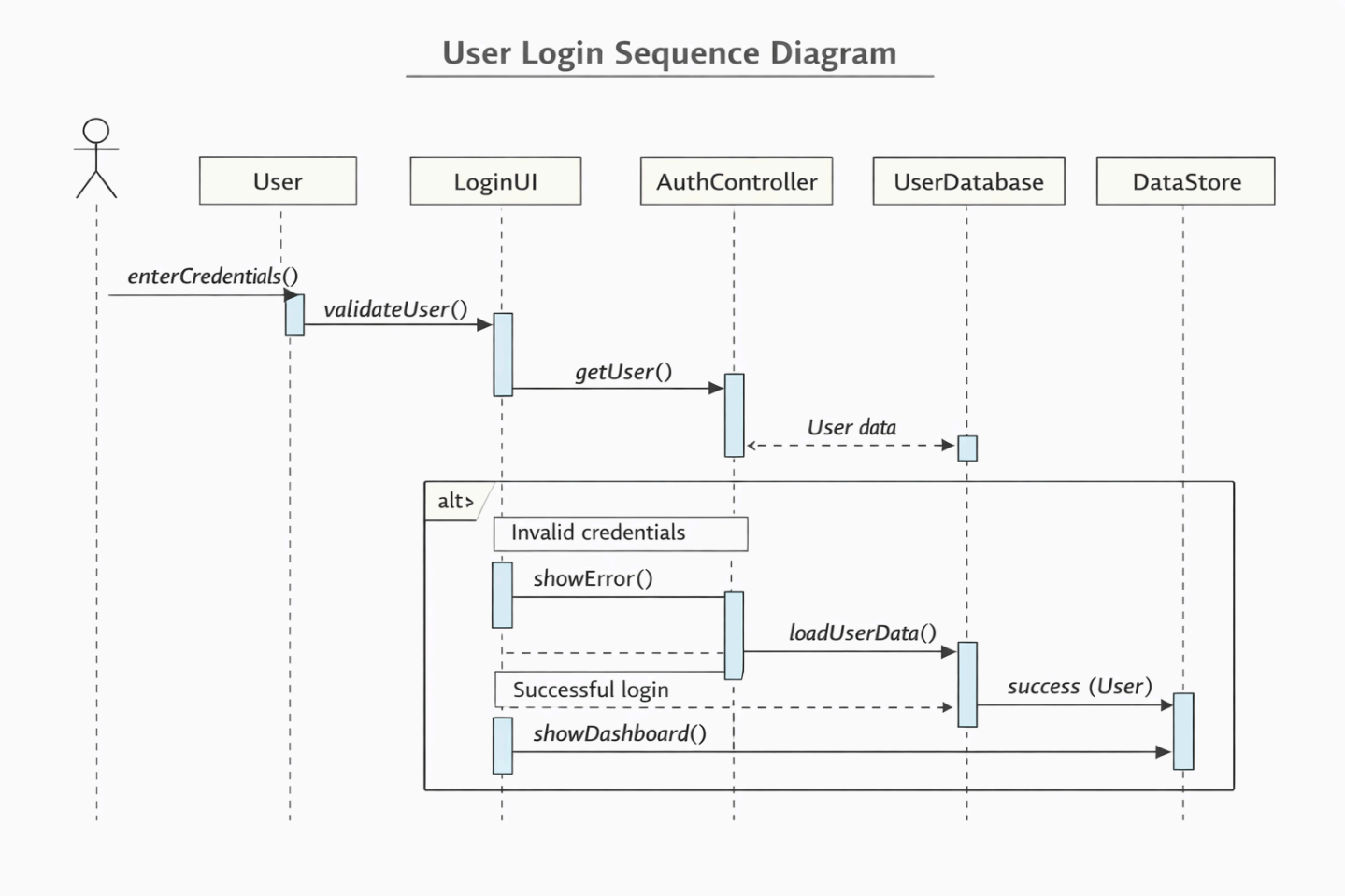
Figure 2: Class Diagram of GloboGym



The class diagram represents the static structure of the system. Core classes include User, Member, Coach, Manager, Membership, Schedule, and Payment. Inheritance is used to model different user roles, while associations define relationships between users, schedules, and payments.

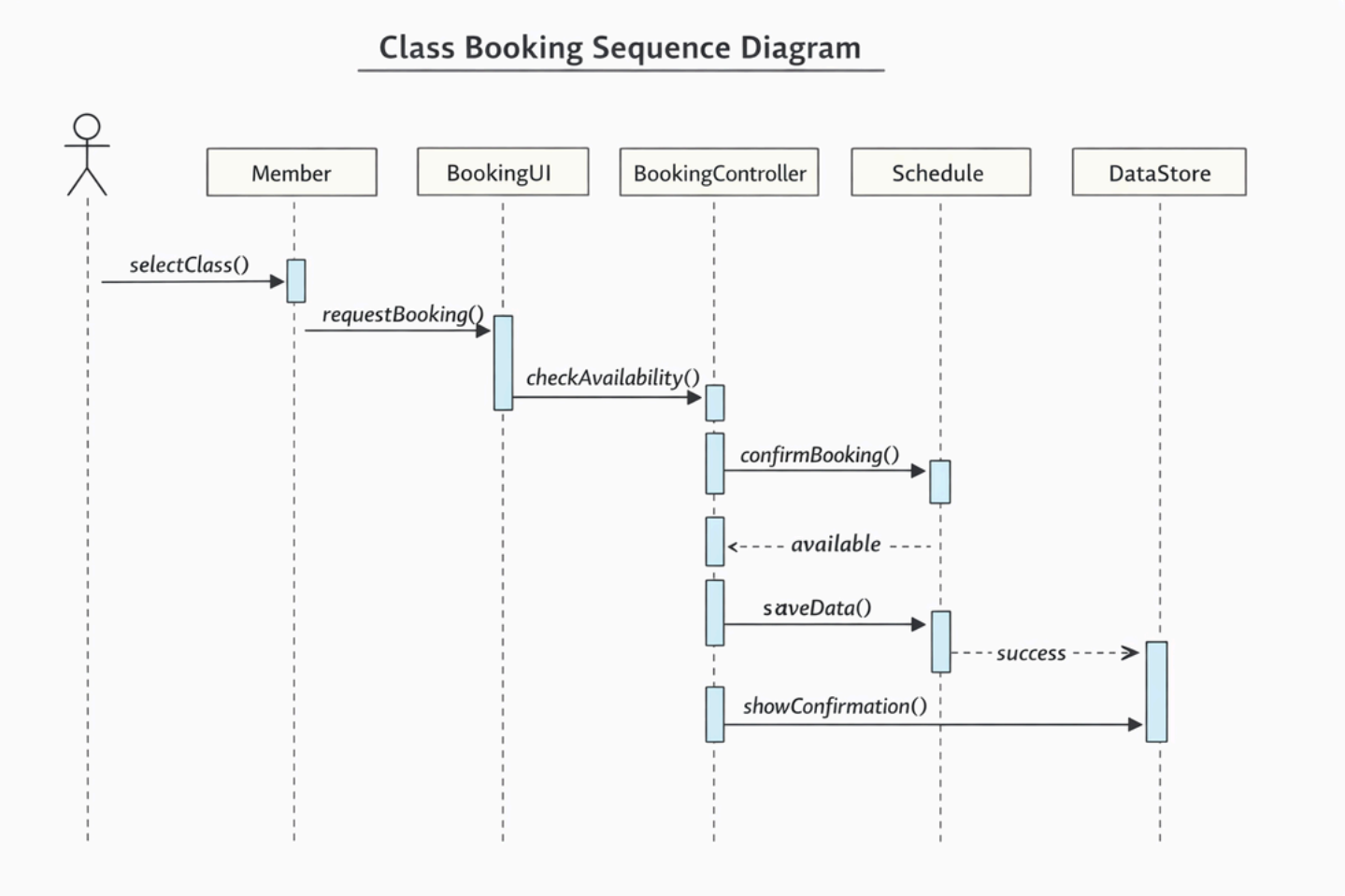
5.3 Sequence Diagrams

Figure 3: Sequence Diagram – User Login Process



This sequence diagram shows the interaction between the user interface, authentication controller, and user data during the login process. It demonstrates how credentials are validated and how access is granted based on user roles.

Figure 4: Sequence Diagram – Class Booking Process



The booking sequence diagram illustrates how a member selects a class, checks availability, confirms booking, and updates the schedule while ensuring data persistence.

6. Technologies Used

- **Programming Language:** Java 22
- **UI Framework:** JavaFX
- **Build Tool:** Maven
- **Persistence:** Java Object Serialization
- **Version Control:** Git & GitHub

7. Implementation Details

7.1 User Interface Implementation

The user interface is developed using JavaFX components such as buttons, forms, tables, and calendar views. Each user role is presented with a customized interface to enhance usability and prevent unauthorized actions.

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

Change picture

UID 6

Username: p4zd4n

Email: p4zd4n@gmail.com

First name: Wiktor

Last name: Chudy

Birth date: 01.08.2003

Update data

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

Username:	p4zd4n
Email:	p4zd4n@gmail.com
Password:	●●●●●●●●
Confirm password:	●●●●●●●●
First name:	Wiktor
Last name:	Chudy
Birth date:	8/1/2003

Update

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

Not purchased

GloboGym

UID 6
Wiktor Chudy

Buy membership! (150.0 PLN = +1 month)

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

Balance: 170.0 PLN

Deposit

2024-06-27: -150.0 PLN
2024-06-27: 20.0 PLN
2024-06-27: 100.0 PLN
2024-06-27: 200.0 PLN

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

Active

Purchased on 2024-06-27

Expires on 2024-07-27, 30 days left

GloboGym

UID 6
Wiktor Chudy

Extend membership! (150.0 PLN = +1 month)

GloboGym

p4zd4n

Logout

Home

Payments

Membership Card

Schedule

Statistics

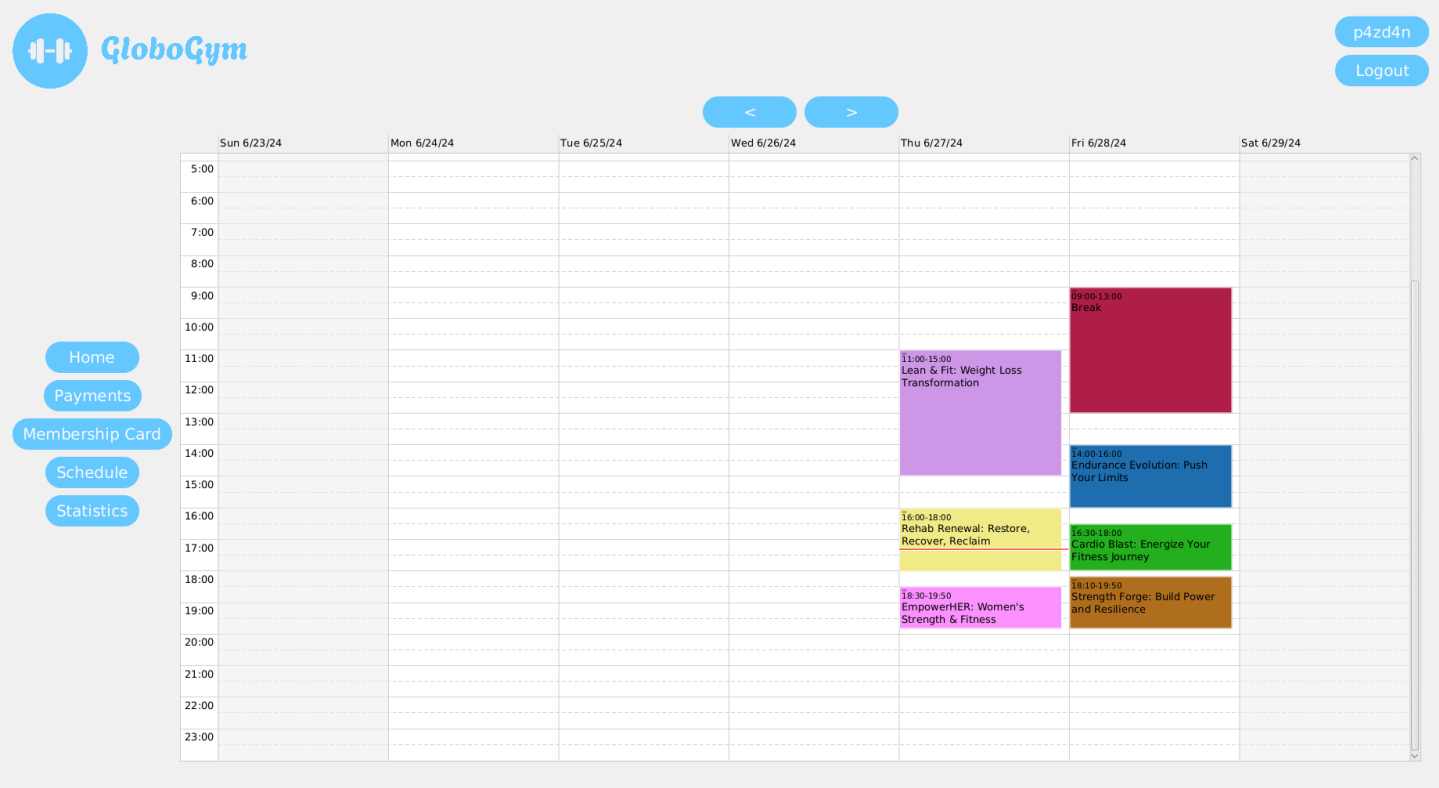
Cardio Blast: Energize Your Fitness Journey details

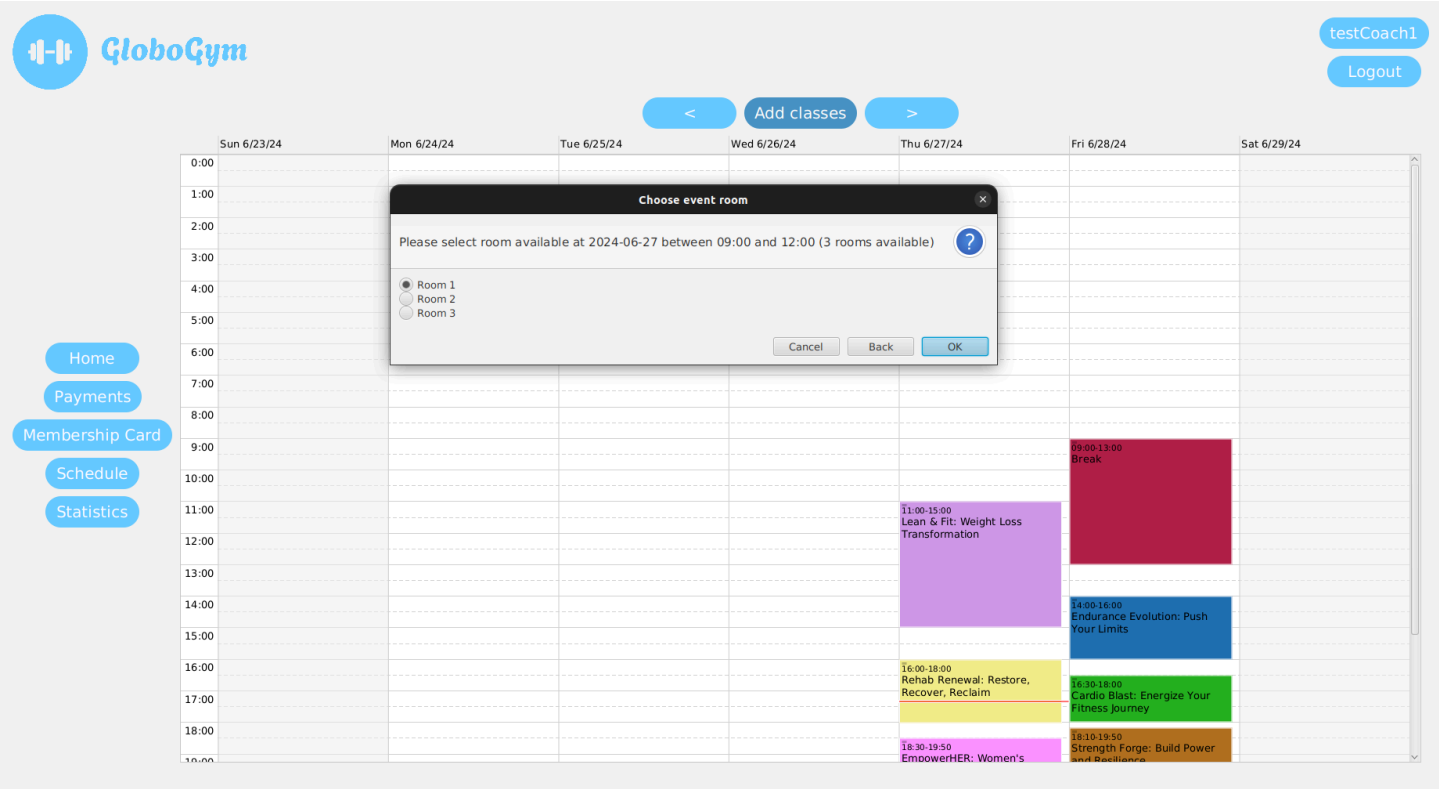
Cardio Blast classes are your ultimate gateway to elevated cardiovascular endurance and overall fitness. These exhilarating sessions are crafted to invigorate your heart rate and ignite calorie-burning potential through a variety of high-energy exercises. Led by enthusiastic instructors, each class incorporates dynamic movements such as running, cycling, jumping jacks, and dance-inspired routines to keep you engaged and motivated. Whether you're a beginner or seasoned athlete, Cardio Blast offers scalable intensity levels to accommodate all fitness levels. Experience the thrill of pushing your limits and improving your cardiovascular health in a supportive group atmosphere that celebrates progress and achievement. Join us for Cardio Blast and unleash your full fitness potential today!

Classes type:	Cardio
Coach:	Eryk Wilk
Start Date/Time:	2024-06-28 16:30
End Date/Time:	2024-06-28 18:00
Room:	3
Free places:	25

Sign Up | Sign Out | OK

	Sun 6/23/24	Mon 6/24/24			Thu 6/27/24	Fri 6/28/24	Sat 6/29/24
5:00							
6:00							
7:00							
8:00							
9:00						09:00-13:00 Break	
10:00							
11:00					11:00-15:00 Lean & Fit: Weight Loss Transformation		
12:00							
13:00							
14:00						14:00-16:00 Endurance Evolution: Push Your Limits	
15:00							
16:00					16:00-18:00 Rehab Renewal: Restore, Recover, Reclaim		
17:00						16:30-18:00 Cardio Blast: Energize Your Fitness Journey	
18:00							
19:00					18:30-19:50 EmpowerHER: Women's Strength & Fitness	18:10-19:50 Strength Forge: Build Power and Resilience	
20:00							
21:00							
22:00							
23:00							


GloboGym
p4zd4n
Logout
<
>
Sun 6/23/24
Mon 6/24/24
Tue 6/25/24
Wed 6/26/24
Thu 6/27/24
Fri 6/28/24
Sat 6/29/24
Home
Payments
Membership Card
Schedule
Statistics
11:00-15:00
Lean & Fit: Weight Loss Transformation
09:00-13:00
Break
14:00-16:00
Endurance Evolution: Push Your Limits
16:00-18:00
Rehab Renewal: Restore, Recover, Reclaim
16:30-18:00
Cardio Blast: Energize Your Fitness Journey
18:10-19:50
Strength Forge: Build Power and Resilience
18:30-19:50
EmpowerHER: Women's Strength & Fitness


GloboGym
testCoach1
Logout
<
Add classes
>
Sun 6/23/24
Mon 6/24/24
Tue 6/25/24
Wed 6/26/24
Thu 6/27/24
Fri 6/28/24
Sat 6/29/24
Choose event room
Please select room available at 2024-06-27 between 09:00 and 12:00 (3 rooms available)
Room 1
Room 2
Room 3
Cancel
Back
OK
Home
Payments
Membership Card
Schedule
Statistics
11:00-15:00
Lean & Fit: Weight Loss Transformation
09:00-13:00
Break
14:00-16:00
Endurance Evolution: Push Your Limits
16:00-18:00
Rehab Renewal: Restore, Recover, Reclaim
16:30-18:00
Cardio Blast: Energize Your Fitness Journey
18:10-19:50
Strength Forge: Build Power
18:30-19:50

GloboGym

testCoach1

Logout

Home

Payments

Membership Card

Schedule

Statistics

Change picture

UID 2

Username: testCoach1

Email: testCoach1@gmail.com

First name: Robert

Last name: Mocny

Birth date: 01.08.2003

Update data

Specializes in:

- [ ] Strength
- [ ] Cardio
- [x] Endurance
- [x] Weight Loss
- [x] Rehabilitation
- [ ] Women Training

GloboGym

testMan

Logout

Add room

Find room

Home

Users Management

Events Management

Rooms Management

Schedule

Statistics

Number	Capacity		
1	10	Update	Remove
2	20	Update	Remove
3	25	Update	Remove
4	16	Update	Remove

You can paste the image from the clipboard.

GloboGym

testMan

Logout

Add classes

Add other event

Find event

Home

Users Management

Events Management

Rooms Management

Schedule

Statistics

ID	Type	Classes type	Name	Start date	End date	Coach	Room		
1	Event		Break	2024-06-28 09:00	2024-06-28 13:00				Remove
2	Classes	Endurance	Endurance Evolution: Push Your Limits	2024-06-28 14:00	2024-06-28 16:00	ID: 2, Robert Mocny	1	Participants	Remove
3	Classes	Rehabilitation	Rehab Renewal: Restore, Recover, Reclaim	2024-06-27 16:00	2024-06-27 18:00	ID: 2, Robert Mocny	2		Remove
4	Classes	Weight Loss	Lean & Fit: Weight Loss Transformation	2024-06-27 11:00	2024-06-27 15:00	ID: 2, Robert Mocny	4		Remove
5	Classes	Cardio	Cardio Blast: Energize Your Fitness Journey	2024-06-28 16:30	2024-06-28 18:00	ID: 3, Eryk Wilk	3	Participants	Remove
6	Classes	Women Training	EmpowerHER: Women's Strength & Fitness	2024-06-27 18:30	2024-06-27 19:50	ID: 3, Eryk Wilk	1	Participants	Remove

Password must consist of lowercase and uppercase letters, numbers and special characters!

Club member Coach (approval required!)

Username: p4zd4n

Email: p4zd4n@gmail.com

Password: ●●●●●

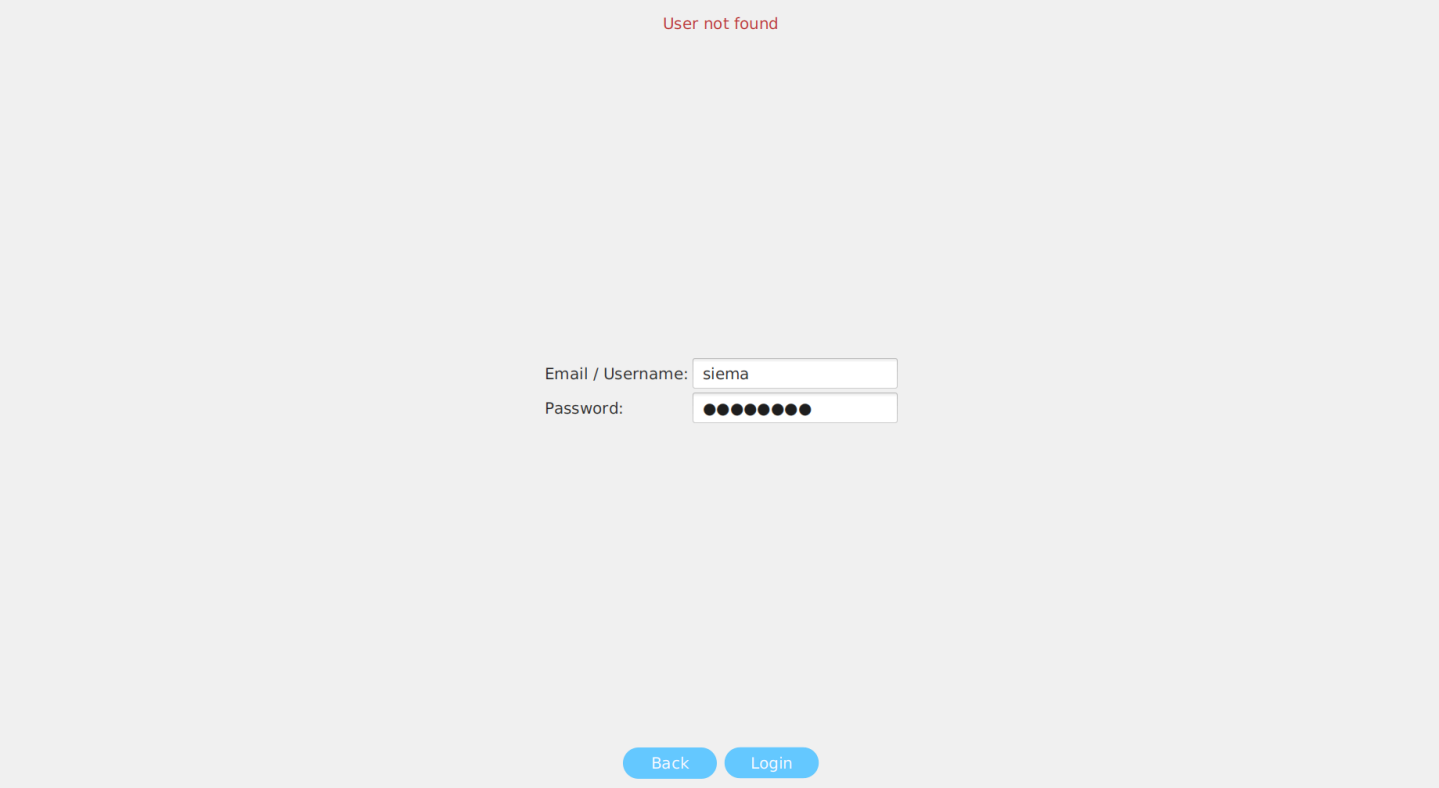
Confirm password: ●●●●●

First name: Wiktor

Last name: Chudy

Birth date: 8/1/2003

Back Register



7.2 Business Logic

The business logic handles:

- User authentication
- Membership validation
- Scheduling constraints
- Payment calculations

Object-oriented concepts such as encapsulation, inheritance, and polymorphism are applied throughout the codebase.

7.3 Data Persistence

Java serialization is used to store application data locally. This ensures that user accounts, schedules, and transactions are preserved between application sessions.

8. Testing and Validation

Testing was conducted manually and focused on:

- User registration and login
- Role-based access validation
- Schedule conflict handling
- Data persistence after application restart

All critical workflows were tested successfully, and no data loss was observed.

9. Challenges and Solutions

Challenge: Managing complex object relationships during serialization
Solution: Structured class design and controlled object references

Challenge: Designing intuitive interfaces for multiple user roles
Solution: Modular UI layouts and consistent navigation patterns

10. Future Enhancements

Potential improvements include:

- Replacing serialization with a relational database (e.g., SQLite)
- Adding report generation (PDF/CSV)
- Implementing unit testing using JUnit
- Introducing cloud-based data synchronization

12. How to Run the project : you should have maven installed in your pc

step 1 : **Using Maven (Recommended)**

mvn clean javafx:run

step 2: **Compile then Run**

mvn clean compile

mvn javafx:run

step 3 : **Package and Run**

mvn clean package

mvn javafx:run

then it should start running please see the readme file for user credentials

11. Conclusion

GloboGym is a complete and functional gym management desktop application developed using Java and JavaFX. The system fulfills all functional requirements, demonstrates effective use of UML modeling, and applies sound software engineering principles. The project provides a strong foundation for future enhancements and real-world deployment.